Chapter 2: Default

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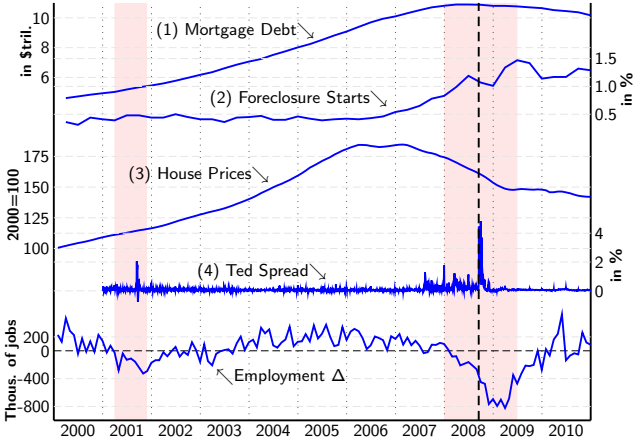
Course Overview



- (1) Debt Growth
- (2) Default
- (3) House Prices
- (4) Financial Crisis
- (5) Institutions
- (6) Expectations

Blinder-Oaxaca decomposition

- $y_2 = \beta_2 x_2$, $y_1 = \beta_1 x_1$
- What explains change in y

$$
\begin{aligned}
y_2 - y_1 &= \beta_2(x_2 - x_1) + x_1(\beta_2 - \beta_1) \\
y_2 - y_1 &= \beta_1(x_2 - x_1) + x_2(\beta_2 - \beta_1) \\
y_2 - y_1 &= \underbrace{\overline{\beta}(x_2 - x_1)}_{\text{(1) Change in } x\text{'s}} + \underbrace{\overline{x}(\beta_2 - \beta_1)}_{\text{(2) Change in } \beta\text{'s}}
\end{aligned}
$$

- used to study race and income
  - (1) is different observables
  - (2) is racism
- For a large sample of loans, share of loans > 60 days delinquent went from 1.2% in 2005 to 8.6% in 2009.

- Reckless lending?
  - Reduced documentation?
  - Adjustable rates?
  - Higher loan size?
  - Lower Credit scores?
- Use B-O decomp:

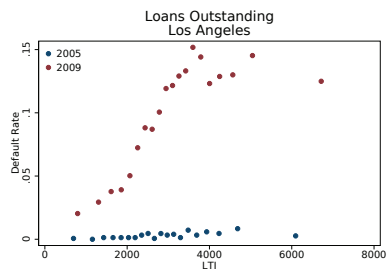
60+DQ in 2009	8.6%
−60+DQ in 2005	1.2%
=Difference	7.4%
=Endowments (x's)	-0.4%
+Coefficients (β's)	7.5%
+Interaction	-0.4%

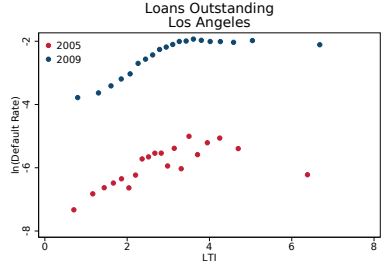
Binscatters

- Focus on *LTI* – Loan Size to Income ratio
- Distribution of x's didn't change that much
- Coefficients
- Do we expect default rates to go up proportionally or linearly?

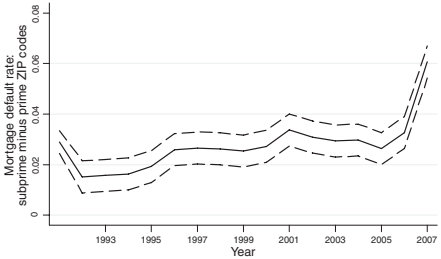
	Borrower Share	2005 defaults Rate	2005 defaults Share	2009 (linear) Rate	2009 (linear) Share	2009 (prop.) Rate	2009 (prop.) Share
Low	50%	0.1%	9%	5.05%	46%	1%	9%
High	50%	1%	91%	5.95%	54%	10%	91%
All		0.55%		5.5%		5.5%	

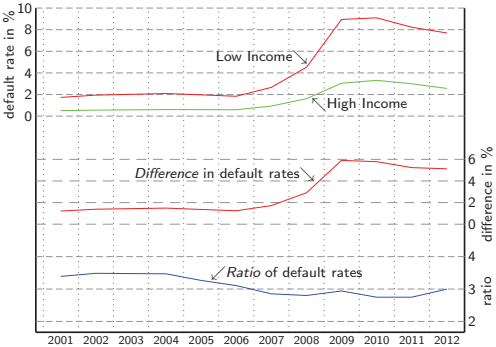
- Why did "coefficients" go up across the board?





Evolution of foreclosure rates





- MS 2009: higher foreclosure growth in low income
- Assume that default rates "should" go up linearly.

Simple version of Epperson et al. (1985)

One-good financial markets model

- 3 periods: t=0,1,2
- States of nature at time 2 $S = \{1, ..., S\}$
- Probability measure P defined over S
- Stochastic house price process H_0, H_1, H_2.
- Riskless asset r
- Absence of arbitrage implies risk-neutral probability Q

$$H_1 = \frac{\mathrm{E}^Q(H_2)}{1+r}$$

Household owns house worth H_0 and borrows L at time 0 with a mortgage with note rate m

- $t = 1$ has three options
  - pay mL dollars
  - sell house for H_1 and replay $(1+m)L$
  - default
- $t = 2$ has two options
  - repay $(1+m)L$ dollars
  - default

Recourse

- Can lender pursue borrower's other assets?
- Ghent and Kudlyak (2011)
- Yes.
- But they almost never do.
- According to GSE Inspector General (2012)

  In 2011, the [GSEs'] vendors pursued 35,231 deficiency accounts, with a combined value of about $2.1 billion. Of this amount, vendors recouped approximately $4.7 million about 0.22%.

Table 1
State foreclosure laws

State	Judicial or Non -Judicial Foreclosure	Optimum Timeline*	Recourse Classification	State	Judicial or Non -Judicial Foreclosure	Optimum Timeline*	Recourse Classification
Alabama	NJ	49–74	Recourse	Nebraska	NJ	121	Recourse
Alaska	NJ	108–111	Non-Recourse	Nebraska	J	176	
Arizona	NJ	115	Non-Recourse	Nevada	NJ	116	Recourse
Arkansas	NJ	90	Recourse	New Hampshire	NJ	75	Recourse
California	NJ	120	Non-Recourse	New Jersey	J	295	Recourse
Colorado	NJ	173	Recourse	New Mexico	J	225	Recourse
Connecticut	J, strict	160	Recourse	New York (NYC)	J	445	Recourse
Connecticut	J, by decree of sale	235		New York (Outside NYC)	J	299	
DC	NJ	48	Recourse	New York (Outside NYC)	NJ	355	
Delaware	J	200–300	Recourse	North Carolina Purchase Mortgages	NJ	120	Non-Recourse
Florida	J	150	Recourse	North Carolina Other Mortgages	NJ	120	Recourse
Georgia	NJ	48	Recourse	North Dakota	J	150	Non-Recourse
Hawaii	NJ	195	Recourse	Ohio	J	217	Recourse
Hawaii	J	320		Oklahoma	NJ	201	Recourse
Idaho	NJ	150	Recourse	Oregon	NJ	160	Non-Recourse
Illinois	J	345	Recourse	Pennsylvania	J	300	Recourse
Indiana	J	266	Recourse	Rhode Island	NJ	74	Recourse
Iowa	J	180	Non-Recourse	South Carolina	J	180	Recourse
Kansas	J	230	Recourse	South Dakota	J	340	Recourse
Kentucky	J	198	Recourse	Tennessee	NJ	50–55	Recourse
Louisiana	J, executory process	209	Recourse	Texas	NJ	35–60	Recourse
Louisiana	J, non-executory	269		Texas	J	80–180	
Maine	J	270	Recourse	Utah	NJ	139	Recourse
Maryland	J	46	Recourse	Vermont	J	275	Recourse
Massachusetts	J	75	Recourse	Virginia	NJ	60	Recourse
Michigan	NJ	360**	Recourse	Washington	NJ	140–150	Non-Recourse
Minnesota	NJ	270–280***	Non-Recourse	West Virginia	NJ	120	Recourse
Missouri	NJ	61–65	Recourse	Wisconsin	J	315	Non-Recourse
Montana	NJ	163	Non-Recourse	Wyoming	NJ	180	Recourse
Mississippi	NJ	90	Recourse				

*These are optimum timelines from the National Mortgage Servicer's Reference Directory, 21st edition (2004). The optimum timelines assume no delays and are based on uncontested foreclosure actions. **The non-judicial foreclosure optimally takes 60 days. However, after that, the redemption period begins to run, typically for six months. Estimated time for completion for uncontested foreclosure without eviction action is 12 months. ***The sale in non-judicial foreclosure can generally be held within 90 days. However, there are substantial redemption rights in Minnesota. Thus, including the redemption period, the optimum timeframe for non-judicial foreclosure is 270–280 days.

The Review of Financial Studies / v 24 n 10 2011

Solution

At t=2

Outcome	Action	Payoff
$H_2 \geq (1+m)L$	Repay	$H_2 - (1+m)L$
$H_2 < (1+m)L$	Default	0

- Payoff $C_2 = \max(H_2 - (1+m)L, 0)$
  - Call option on the house with strike price equal to the unpaid principal balance
  - Absence of arbitrage $\Rightarrow C_1 = \mathrm{E}^Q(C_2)$

At t=1. Assume $H_1 < (1+m)L$ so no sale

Outcome	Action	Payoff
$C_1 \geq mL$	Repay	$C_1 - mL$
$C_1 < mL$	Default	0

- Payoff $M_1 = \max(C_1 - mL, 0)$
  - Call option on the call option with strike price equal to the monthly payment
  - Absence of arbitrage $\Rightarrow M_0 = \mathrm{E}^Q(M_1)$

- Getting a mortgage is equivalent to:
  - Selling house to the lender for a "compound option":
  - Option at $t = 1$ to buy option for mL to buy house at $t = 2$ for $(1+m)L$.

Three equivalent formulations at $t = 1$

(C) Call Option formulation

$$\text{Default} \Leftrightarrow mL > C_1 = \frac{1}{1+r}\mathrm{E}^Q(\max(H_2 - (1+m)L, 0))$$

- Put-call parity

$$\max(H_2 - (1+m)L, 0) = H_2 - (1+m)L - \max((1+m)L - H_2, 0)$$

- Implies that:

$$\begin{aligned} C_1 &= \tfrac{1}{1+r}\mathrm{E}^Q(H_2 - (1+m)L - \max((1+m)L - H_2, 0)) \\ &= \tfrac{1}{1+r}\mathrm{E}^Q(H_2) + \tfrac{1+m}{1+r}L - \tfrac{1}{1+r}E^Q \max((1+m)L - H_2, 0)) \\ &= H_1 + \tfrac{1+m}{1+r}L - P_1 \end{aligned}$$

(P) Put Option formulation

$$\text{Default} \Leftrightarrow mL > H_1 + \tfrac{1+m}{1+r}L - \tfrac{1}{1+r}E^Q \max((1+m)L - H_2, 0))$$

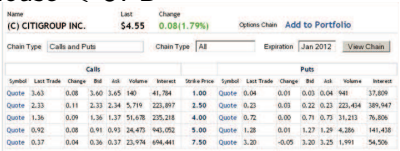
(V) Value formulation

$$\text{Default} \Leftrightarrow \underbrace{mL + \tfrac{1+m}{1+r}L - P_1}_{\text{Value of Mortgage}} > \underbrace{H_1}_{\text{Value of House}}$$

Mortgage as a call option

$$\text{Default} \Leftrightarrow mL > C_1 = \frac{1}{1+r}\mathrm{E}^Q(\max(H_2 - (1+m)L, 0))$$

- Borrower sells house to lender
  - Option to repurchase the house for UPB $\Rightarrow$ prepayment.
  - Option to buy another option $\Rightarrow$ make monthly payment.
  - A borrower defaults by *not* exercising the options.
- What about negative equity? value of house $<$ UPB

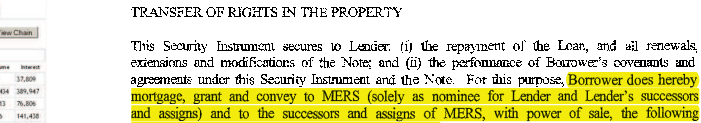


Name: (C) CITIGROUP INC. Last: $4.55 Change: 0.08(1.79%) Options Chain Add to Portfolio

Chain Type: Calls and Puts Chain Type: All Expiration: Jan 2012 View Chain

Calls								Puts							
Symbol	Last Trade	Change	Bid	Ask	Volume	Interest	Strike Price	Symbol	Last Trade	Change	Bid	Ask	Volume	Interest
Quote	3.63	0.08	3.60	3.65	140	41,784	1.00	Quote	0.04	0.01	0.03	0.04	941	37,809
Quote	2.33	0.11	2.33	2.34	5,719	223,897	2.50	Quote	0.23	0.03	0.22	0.23	223,454	389,947
Quote	1.36	0.09	1.36	1.37	51,678	235,218	4.00	Quote	0.72	0.00	0.71	0.73	31,213	76,806
Quote	0.92	0.08	0.91	0.93	24,473	943,052	5.00	Quote	1.28	0.01	1.27	1.29	4,286	141,438
Quote	0.37	0.04	0.36	0.37	23,974	694,441	7.50	Quote	3.20	-0.05	3.20	3.25	1,991	54,506

- Common law mortgage – ancient
- "Right of equitable redemption" – right to repurchase house
- To take away house, lender needs to "foreclose" right of redemption.
- Still the law in 28 states



TRANSFER OF RIGHTS IN THE PROPERTY

This Security Instrument secures to Lender: (i) the repayment of the Loan, and all renewals, extensions and modifications of the Note; and (ii) the performance of Borrower's covenants and agreements under this Security Instrument and the Note. For this purpose, Borrower does hereby mortgage, grant and convey to MERS (solely as nominee for Lender and Lender's successors and assigns) and to the successors and assigns of MERS, with power of sale, the following described property located in the county of NORFOLK :

Default as a put option

$$\text{Default} \Leftrightarrow mL > H_1 + \frac{1+m}{1+r}L - \frac{1}{1+r}\mathrm{E}^Q \max((1+m)L - H_2, 0)$$

- Homeowner
  - Borrows money
  - Plus "put:" option to sell the house to the lender for UPB.
- Assume $m = r = 0$

Assets	Liabilities
House ($\mathrm{E}^Q(H_2)$)	Loan (L)
Put Option ($\mathrm{E}^Q \max(L - H_2, 0)$)	

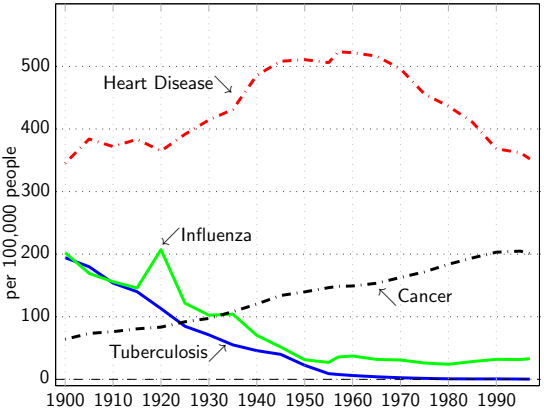
$$\text{Net Worth} = \mathrm{E}^Q \max(L, H_2)$$

- Leverage and house prices
  - Low leverage $\Rightarrow \partial NW/\partial H = 1$
  - High leverage prices $\Rightarrow \partial NW/\partial H = 0$
- Put option is insurance
  - Defaults are a sign of risk shareing
  - Dubey, Geanakoplos, and Shubik (2005), Zame (1993)

The Double Trigger Model

Life Event? \ Negative Equity?	Yes	No
Yes	Default	Repay
No	Repay	Repay

Survival Analysis

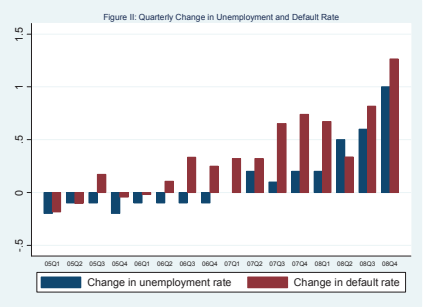


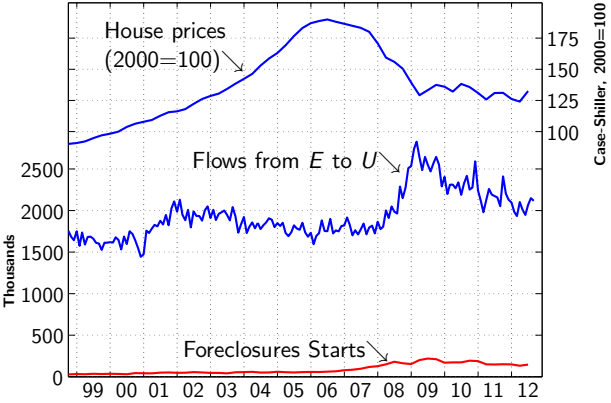
- Medical science largely conquered infectious diseases
- Reduction in "competing risks"
- Heart disease and cancer
- Same issue with mortgages
  - Default
  - Prepayment
- In OFHEO stress test, there were two *interest rate* scenarios for fixed rate mortgages

Scenario	Default %	Prepay %
Up rate	2.35%	31.50%
Down rate	1.42%	93.87%

Employment and default

While the change in the default rate turns positive in the second quarter of 2006, the unemployment rate does not increase until the second quarter of 2007.
In short the aggregate pattern in Figure II is not consistent with the view that unemployment shock are the primary drivers of increasing default rates. [emphasis added] (Mian, 2010)





Guiso, Sapienza, and Zingales (2013) and Strategic Default

- Strategic default occurs when borrowers, "choose to walk away from their houses even if they can afford to pay their mortgages."
- Survey data
- Credit bureau data: Prioritizing debt
  - Keys et al. (2012), Tirpattur, Chang, and Egan (2010), Experian and Oliver Wyman (2009)

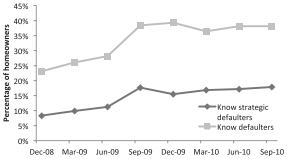
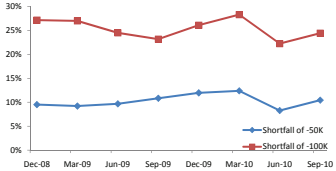


Figure 5. Percentage of people who know a defaulter. "Know defaulters" is the percentage of homeowners who report they know at least one person who has defaulted on his/her mortgage when answering the question "How many people do you know who have defaulted on their house mortgage?" "Know strategic defaulters" is the percentage of homeowners who know at least one strategic defaulter based on the answers to the question: "Of the people you know who have defaulted on their mortgage, how many do you think walked away even if they could afford to pay the monthly mortgage?"

Figure 1. Percentage of people willing to default strategically

n of homeowners who say they are willing to default when the value of their home equity falls short of the value of the
50K and 100K respectively even if the homeowner can afford to pay the monthly mortgage costs.



A. Percentage defaulting and relative shortfall

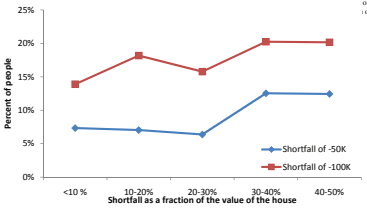
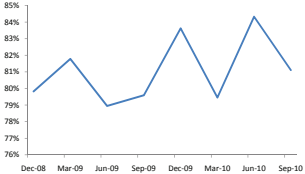


Figure 3. Fraction of people who think it is morally wrong to walk away

of people responding positively to the following question: "Do you think that it is morally wrong to walk away from a
one can afford to pay the monthly mortgage?"



Gerardi et al. (2015)

- 2009-2013 PSID Mortgage Distress Supplement to role of strategic factors and shocks in default.
  - Self-reported home value, principal, months delinquent.
  - Job loss information for both head and spouse, wealth, and income.
  - Complete consumption expenditure data since 2005 (comparable to CEX).

Measuring Strategic Default

- Simple budget constraint (ignoring assets).

$$c + m \leq y$$

- **residual income**$= y - m$.
- Two questions
  1. Is m unaffordable? Is c above subsistence?

     $$y - m < c_{VA} \Rightarrow \text{Unaffordability}$$

  2. Is m affordable? Does household need to cut c?

     $$y - m > c_{t-1} \Rightarrow \text{Affordability}.$$

- **How do we measure subsistence consumption?** use VA minimum residual income thresholds to define c_{VA} (depends on family size / location).

Residual Income Requirements For VA Home Loans

For loan Amounts $79,999 and below, the residual income must exceed:

Family Size	Northeast	Midwest	South	West
1	$390	$382	$382	$425
2	$654	$641	$641	$713
3	$788	$772	$772	$859
4	$888	$868	$868	$967
5	$921	$902	$902	$1,004
Over 5	+ $75 per person up to a family of 7	+ $75 per person up to a family of 7	+ $75 per person up to a family of 7	+ $75 per person up to a family of 7

Low residual income is highly correlated with adverse shocks

Table : Adverse shock incidence by residual income quintile.

	Quintile of Residual Income (Gross Income-Mortgage Expenses)				
	1	2	3	4	5
30% Income Drop or more (d)	25.6%	12.2%	5.9%	4.7%	3.9%
Unemployed Head Last Year (d)	13.9%	7.1%	4.9%	4.0%	4.2%
Unemployed Spouse Last Year (d)	4.3%	6.1%	5.7%	3.2%	3.1%
Transition into Disability, Head (d)	6.0%	4.3%	3.7%	3.4%	2.7%
Transition into Disability, Spouse (d)	3.4%	3.9%	3.4%	3.6%	3.8%
Any Shock (Unemployment, Disability, or 30% Income Drop) (d)	40.9%	28.5%	20.0%	16.8%	16.1%

Notes: This table displays the fraction unemployed, recently disabled, and with a 30% income drop or more, split by quintile of residual income (defined as gross family income less mortgage expenses) for the sample of PSID households surveyed in 2009, 2011, and 2013.

Results

	(1) "Can Pay" $c < y - m$ # (i)	share (ii)=(i)/(vii)	(2) $c > y - m > c(VA)$ # (iii)	share (iv)=(iii)/(vii)	(3) "Can't Pay" $y - m < c(VA)$ # (v)	share (vi)=(v)/(vii)	(4) Total # (vii)
A. Total							
Default	74	0.377	65	0.333	57	0.291	196
Population	5,173	0.699	1,704	0.230	531	0.072	7404
Default Rate (# Def./# Pop.)		0.014		0.038		0.107	0.027
B. $LTV > 90$							
Default	47	0.409	41	0.352	28	0.239	115
Population	1,117	0.664	428	0.254	140	0.083	1684
Default Rate		**0.042**		0.095		**0.197**	0.069
C. $LTV < 90$							
Default	27	0.330	25	0.306	29	0.364	81
Population	4,056	0.709	1,277	0.223	391	0.068	5720
Default Rate		**0.007**		0.019		**0.075**	0.014

1. ($\nearrow$) shocks matter
2. ($\leftrightarrow$): many defaults are "strategic" 41% of high LTV defaults
3. ($\leftrightarrow$): many defaulters cannot afford payments
4. ($\updownarrow$) strategic default is really rare
5. ($\updownarrow$) "can't pay" default is rare

Hsu, Matsa, and Melzer (2014)

- Use state level variation in unemployment benefits to measure the effect of cash flow on default
- Bigger benefits are associated with fewer defaults
- Interaction with layoff
- Conclude that expanded UI during the crisis prevented 1.4 million foreclosures.

Table II: Unemployment Insurance Generosity and Mortgage Delinquency (SIPP, 1991-2011)

	Dependent variable: Mortgage Delinquency		
	(1)	(2)	(3)
Max Benefit	-0.08	-0.03	
	(0.12)	(0.13)	
Max Benefit*Layoff		-0.23***	-0.25***
		(0.07)	(0.08)
Layoff	6.51***	6.55***	6.55***
	(0.38)	(0.34)	(0.35)
Observations	64,922	64,922	64,922
R^2	0.05	0.05	0.05
Household-level controls?	Y	Y	Y
State-year controls?	Y	Y	-
State and year FEs?	Y	Y	-
State-year FEs?	N	N	Y

Notes: This table summarizes the results from linear probability regressions of mortgage delinquency on *Max Benefit* (the maximum total potential benefit available under the state's unemployment insurance system), a layoff indicator, their interaction, and a set of controls. When uninteracted, the *Layoff* coefficient measures the effect in a state with average *Max Benefit*. Controls in all regressions include the household's earnings, net worth, education (indicators for each of five categories), mortgage loan-to-value, and indicators for negative home equity and its interaction with *Layoff*, and the states' unemployment rate, average wage, GDP growth rate, home price growth rate, Unemployment Insurance (UI) Trust Fund reserve ratio, an indicator for a negative UI Trust Fund reserve ratio, and state and year fixed effects. In column (3), controls also include state-by-year fixed effects, which absorbs the estimate of the *Max Benefit* main effect. Standard errors, adjusted for clustering at the state level, are reported in parentheses.
* significant at 10%; ** significant at 5%; *** significant at 1%

Table VII: Delinquencies and Foreclosures Avoided by Unemployment Insurance Expansions (2008-2012)

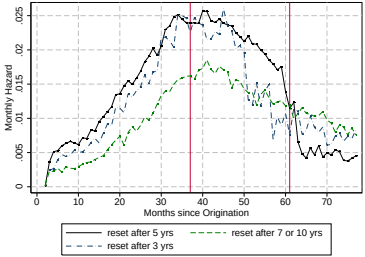
Year	Pr(Layoff)	UI-Layoff Coefficient	EB, EUC & FAC	Pr(Deliq.)	Delinquent Loans (D)	Foreclosure Starts (F)	ΔD	ΔF
2008	0.174	-0.31	5,452	7.65%	3,315,069	951,090	-127,432	-36,560
2009	0.186	-0.31	18,988	7.65%	4,112,037	2,320,315	-588,494	-332,072
2010	0.176	-0.31	30,135	7.65%	4,031,198	2,132,809	-866,412	-458,398
2011	0.163	-0.31	28,281	7.65%	3,465,668	1,780,284	-647,403	-332,565
2012	0.163	-0.31	22,528	7.65%	3,058,442	1,475,146	-455,107	-219,507
						TOTAL	-2,684,848	-1,379,102

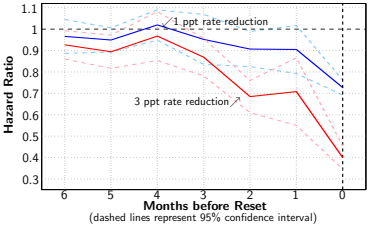
Fuster and Willen (forthcoming)

- Default condition:

$$mL > C_1 = \frac{1}{1+r}\mathrm{E}^Q(\max(H_2 - (1+m)L, 0))$$

- Consider adjustable rate mortgages whose payment went down
  - 3/1 ARM (month 36)
  - 5/1 ARM (month 60)
  - 7/1 ARM (month 84)
- 50% reduction in payment leads to a 55% reduction in default hazard



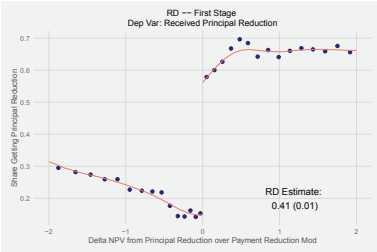


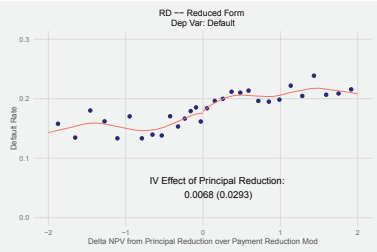
Ganong and Noel (2016)

- Default condition:

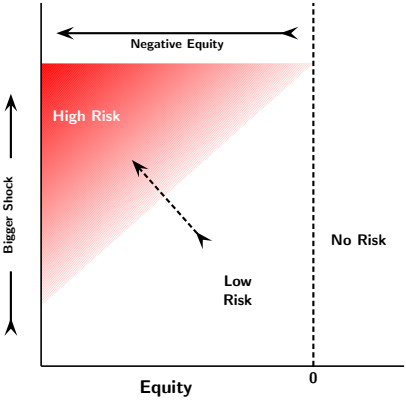
$$mL > C_1 = \frac{1}{1+r}\mathrm{E}^Q(\max(H_2 - (1+m)L, 0))$$

- To prevent default:
  - Lower m
  - Lower L
- Many people demanded reductions in L
  - Major political issue
- Exploit discontinuity in government program – HAMP
- Find that payment reduction is effective





Modifications to the model



- Default condition:
$$mL > C_1 = \frac{1}{1+r}\mathrm{E}^Q(\max(H_2 - (1+m)L, 0))$$
- Tradeoff between
  - Cost of making payment
  - Benefit in the future
- With borrowing constraints $r^h \neq r$
$$\frac{1}{1+r^h} = \frac{\mathrm{E}(u'(c^h_{t+1}))}{u'(c^h_{t+1})}$$
- Models: Foote, Gerardi, and Willen (2008), Schelkle (2011), Laufer (2012), Campbell and Cocco (2011), Gerardi, Shapiro, and Willen (2007)

The default puzzle

- Default models massive over predict defaults
- Kau, Keenan, and Kim (1994): 100 percent of borrowers with LTV ratios above 115 percent will default.
- To get realistic default rates
  - Schelkle (2011): strong aversion of borrowers to renting
  - Laufer (2012): massive utility penalty for defaulting
- In PSID, of people with negative equity *and* with consumption below subsistence
  - 80% made their payments

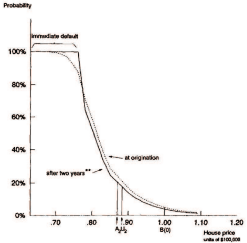


FIG. 2. Probability of default vs house price H. U_2 is the unpaid principal after two years, before payment; A_2 is the present value of payments after two years, before payment. All parameters are at their base values except as indicated in the figure. **The probability of default after two years represents the probability at the second year that thereafter the mortgage defaults.

The slide you've all been waiting for...

- The end.

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